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StayChain

Decentralised Accommodation Booking Protocol on BOT Chain

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PROTOCOL OVERVIEW

StayChain is a decentralised accommodation booking protocol deployed on BOT Chain. Hotels list rooms with nightly rates; guests book specific day ranges by paying the exact total into the contract. Availability is enforced on-chain via a taken[room][day] mapping — double-booking is impossible. Check-in by the hotel atomically releases the full payment. Cancellations more than 2 days before check-in receive a full refund; later cancellations receive a 50% refund. After check-in, guests may leave a 1–5 star on-chain review.

A technical overview of the protocol, its design, core functions, security properties, and roadmap.

1. Problem Statement

• Platform dependency — Airbnb, Booking.com, and similar platforms charge 15–20% commission, retain user data, and can delist hosts without cause.

• Payment timing risk — guests pay upfront but hotels may cancel; hosts wait weeks for payment release.

• Double-booking — without atomic availability checks, the same room may be booked twice.

• Review integrity — centralised reviews can be suppressed, manipulated, or purchased.

• No on-chain proof of stay — there is no verifiable record that a guest stayed at a property.

2. Protocol Design

2.1 Availability Model

Days are expressed as Unix day numbers (timestamp / 86400). The taken[roomId][day] mapping is set to true for each booked day in a single transaction. require(!taken[rid][startDay + i]) ensures strict availability — any overlap with an existing booking reverts the transaction.

2.2 Payment & Cancellation Policy

`Scenario`
`Refund`

Cancel > 2 days before startDay

100% of paid amount

Cancel <= 2 days before startDay

50% of paid amount; 50% to hotel

Check-in (hotel confirms)

100% released to hotel

2.3 Review System

Reviews are on-chain (ratings[bookingId] = 1–5) and can only be submitted by the verified guest after a successful check-in. They cannot be deleted, edited, or suppressed.

3. Contract Architecture

3.1 Data Model

Room: hotel address, name (string), nightly rate (wei).

Booking: roomId, guest address, startDay, nights, paid (wei), checkedIn flag, cancelled flag.

taken mapping: roomId => day => bool.

3.2 Write Functions

`Function`
`Role`
Description

`addRoom(name, nightly)`
Any Hotel

Lists a room at a nightly rate

`book(roomId, startDay, nights)`
Guest

Books and pays — exact ETH = nightly * nights

`cancel(bookingId)`
Guest

Cancels booking (partial/full refund by policy)

`checkIn(bookingId)`
Hotel

Confirms guest arrival — releases payment to hotel

`review(bookingId, 1-5)`
Guest (post check-in)

Submits immutable on-chain rating

4. Security Properties

4.1 Atomic Availability

The loop over nights inside book() either succeeds for all days or reverts for all. No partial booking is possible.

4.2 Payment Custody

Funds are held by the contract until checkIn() is called. The hotel cannot access the payment until the guest is verified as checked in.

4.3 Cancellation Availability Restoration

`cancel() iterates through all booked days and sets taken[room][day] = false, immediately freeing the room` for rebooking.

4.4 Review Integrity

`review() requires bookings[bid].checkedIn — only guests who physically checked in (as confirmed by the` hotel) can leave a review.

5. Events

The public rooms and bookings arrays and the public ratings mapping allow full enumeration by any frontend without custom events.

6. Deployment

`Parameter`
Value

`Chain`
BOT Chain (Chain ID 968)

`RPC`
https://rpc.botchain.ai

Day unit

Unix day = floor(block.timestamp / 86400)

Cancellation policy

>2 days: 100% refund; <=2 days: 50% refund

7. Roadmap

Milestone	Description
v1.1	Configurable per-room cancellation policy
v1.2	ERC-20 stablecoin payments
v1.3	ERC-721 NFT booking confirmation receipts
v2.0	On-chain dispute resolution with arbitration deposit